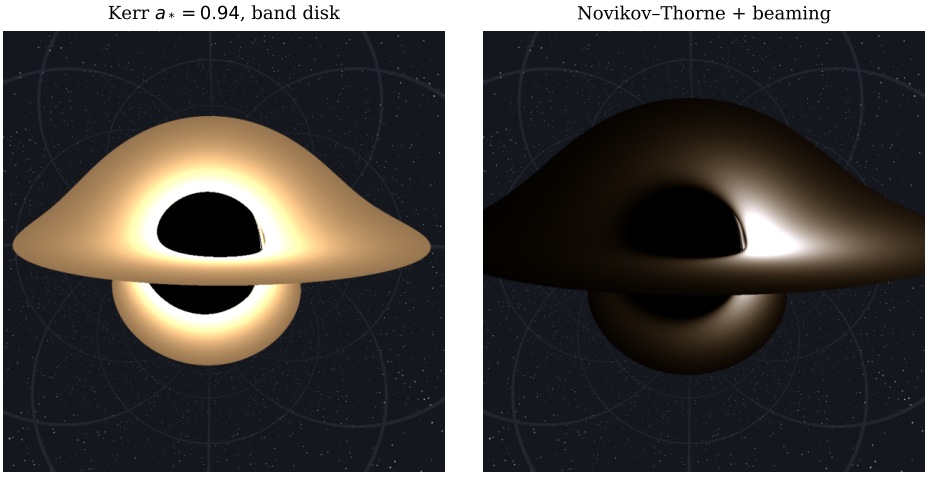
SGR A*

A First-Principles Relativistic Simulator of the Galactic-Centre Black Hole

Shadow metrology, S-star orbit fitting, and a free-quadrupole test of the no-hair theorem — with a complete account of the methods, tests, and development of the instrument



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Abstract

We present a ground-up, physically rigorous simulator of the Galactic-centre black hole Sgr A*, built on exact Kerr geodesics and validated end-to-end against published strong-field observations. Null and timelike geodesics are integrated in a second-order Mino-time formulation of the Carter equations with per-step constraint projection, conserving (E, L_z, Q) to better than 10^{-6} relative over complete orbits. A CPU reference ray tracer reproduces the analytic Schwarzschild shadow to a relative accuracy of 5×10^{-16} and the Bardeen critical curve for spinning holes; a double-precision GPU compute port agrees with the CPU ground truth to a mean pixel residual of 0.00289. Fitting 13 orbital and potential parameters to the 145 astrometric and 44 radial-velocity measurements of the star S2 (Gillessen et al., 2017) with full general-relativistic timelike geodesics recovers $M = 4.270 \pm 0.195 \times 10^6\,\mathrm{M}_\odot$ and $R_0 = 8272 \pm 182$ pc, within 0.14σ and 0.03σ of the GRAVITY interferometric values, at a reduced χ^2 of 0.959. The model reproduces the gravitational-redshift excess at the 2018 pericentre passage ($205\,\mathrm{km\,s^{-1}}$ peak) detected by GRAVITY Collaboration (2018), and the Schwarzschild precession of 11.98 arcmin per orbit confirmed by GRAVITY Collaboration (2020). At the GRAVITY mass and distance the simulated photon ring diameter is $49.68\,\mu$as, inside the EHT measurement of $51.8 \pm 2.3\,\mu$as (Event Horizon Telescope Collaboration, 2022). Freeing the quadrupole moment in a Hartle–Thorne extension yields $\varepsilon_q = 0.00 \pm 0.93 \times 10^5$ (68 per cent), consistent with the no-hair theorem.

This report is the long-form account of that programme. Part I is a plain-language primer for readers without a physics background. Parts II–IV derive, from first principles, every formula the simulator implements; document the numerical methods and their error analysis; present the full results with their statistical treatment; and record the complete development history, test suite, and validation gates through which the instrument was built.

Notation and Conventions

Geometric units $G = c = 1$ are used throughout unless stated; masses, lengths and times are quoted in units of the gravitational radius M (for Sgr A*: $GM/c^2 = 5.1243^{-1}$ scaling on the sky, $GM/c^3 = 21.17$ s). Metric signature $(-,+,+,+)$; Boyer–Lindquist coordinates (t, r, θ, ϕ).

Reproducibility Statement

Every quantitative statement in this report that is not a cited literature value is typeset from a macro in `numbers.tex`, generated by `scripts/make_numbers.py` from machine-readable artefacts produced by the simulation; every figure is generated by `make figures` (matplotlib) from the same artefacts. The single command `make reproduce` re-runs the full chain — unit tests, reference renders, sub-ring metrology, orbit fits, calibration, sweeps, and figure generation — from source. A number that cannot survive this cycle cannot appear in this document.

Part I

Foundations for the Non-Specialist

Chapter 1

Gravity, Mass, and the Idea of a Black Hole

[Chapter under construction — scope: escape velocity, the Schwarzschild radius, horizons, singularities, and a sense of scale. See `paper/THESIS_PLAN.md`.]

Chapter 2

Spacetime and Einstein's Gravity, Gently

[Chapter under construction — scope: curved spacetime, geodesics, light bending, time dilation, and what a metric is. See `paper/THESIS_PLAN.md`.]

Chapter 3

Spinning Black Holes and the Shadow

[Chapter under construction — scope: Kerr vs Schwarzschild, frame dragging, the ergosphere, the photon sphere, and how the shadow forms. See `paper/THESIS_PLAN.md`.]

Chapter 4

Light, Heat, and Colour: Reading the Disk

[Chapter under construction — scope: blackbody colour, the accretion disk, Doppler beaming, and the three redshifts. See `paper/THESIS_PLAN.md`*.]*

Chapter 5

Orbits, S-Stars, and Weighing a Black Hole

[Chapter under construction — scope: Kepler's ellipse, astrometry vs radial velocity, precession, and how mass and distance emerge. See `paper/THESIS_PLAN.md`.]

Chapter 6

The Toolkit in Plain Words

[Chapter under construction — scope: geometric units, tensors gently, chi-squared and confidence intervals, MCMC, and the no-hair theorem in one picture. See `paper/THESIS_PLAN.md`.]

Part II

Theory and Methods

Chapter 7

Introduction and Strong-Field Context

7.1 Sagittarius A*: a dual strong-field laboratory

The compact radio source Sgr A* at the dynamical centre of the Milky Way is the nearest supermassive black hole and the only one for which two independent strong-field probes are available at once: the resolved orbits of the S-star cluster within a few thousand gravitational radii (Gillessen et al., 2017; GRAVITY Collaboration, 2018, 2020, 2022), and the horizon-scale image of its photon ring obtained by the Event Horizon Telescope (Event Horizon Telescope Collaboration, 2022). Together they anchor the mass ($4.297 \pm 0.012 \times 10^6\,\mathrm{M}_\odot$), the distance ($8277 \pm 9$ pc), and the angular size of the emission ring ($51.8 \pm 2.3\,\mu$as), making Sgr A* the cleanest laboratory for testing general relativity (GR) in the strong-field, low-velocity regime.

7.2 The no-hair theorem as a falsifiable target

The no-hair theorem states that the exterior spacetime of an astrophysical black hole is fully specified by its mass M and spin a; in particular the mass quadrupole moment is locked to $Q = -Ma^2$. Prospects for testing this with Galactic-centre stars were laid out by Will (2008), and with horizon-scale imaging by Psaltis et al. (2016). Both programmes require a forward model that is exact in the strong field: post-Newtonian expansions degrade at the S2 pericentre ($r_{\mathrm{peri}} \approx 1470\,R_S$), and the photon ring is intrinsically a property of exact null geodesics (Bardeen, 1973; Johnson et al., 2020).

7.3 This programme

This report describes such a forward model, built from first principles and validated at every layer. The simulator integrates the Carter-separated geodesic equations (Carter, 1968) in Kerr spacetime, renders the accretion flow with the standard relativistic thin-disk model (Novikov & Thorne, 1973; Page & Thorne, 1974), fits real astrometric and spectroscopic data for the star S2, and embeds a free-quadrupole Hartle–Thorne metric (Hartle & Thorne, 1968; Glampedakis & Babak, 2006) to constrain departures from Kerr. The same geodesic core runs on the GPU for interactive exploration, and every number quoted here is regenerated from the code by a single reproduction target.

7.4 Lineage of the imaging problem

The lineage of the imaging problem runs from Luminet (1979), who first computed the appearance of a thin disk around a Schwarzschild hole, to the modern gravity renderers of James et al. (2015); our CPU reference tracer follows the same backward ray-tracing logic but is held to numerical standards set by the analytic theory (Chapter 10).

> *[Chapter under construction — scope: Expanded historical thread — Kerr 1963 → Boyer–Lindquist 1967 → Carter 1968 → Bardeen 1973 → Cunningham–Bardeen 1973 → Luminet 1979 → James et al. (Interstellar) 2015 → GRAVITY 2018/2020/2022 → EHT 2022 — lands with the expanded bibliography. See* `paper/THESIS_PLAN.md`*.]*

7.5 Contributions

The specific contributions of this programme are:

1. a turning-point-free second-order Mino-time integration of the Carter equations with per-step constraint projection, which makes traced shadow boundaries independent of integrator tolerance (Chapter 10);
2. machine-precision validation of the traced Schwarzschild shadow against Synge's formula and of the Kerr shadow against the analytic critical curve (Chapter 11);
3. a 13-parameter full-geodesic fit to the public S2 data that recovers the GRAVITY mass and distance and reproduces both published relativistic effects with no tunable amplitudes (Chapter 15);
4. a quantified no-hair constraint from S2 alone, with a covariance forecast for continued monitoring (Chapter 17);
5. a validated double-precision GPU implementation of the same geodesic core (Chapter 18); and
6. a single-command reproduction pipeline in which every number in this report is regenerated from source (Appendix D).

7.6 Reader's guide

The report is organised in four parts so that readers with different backgrounds can enter at different depths.

Part I — Foundations for the non-specialist assumes no physics background. It explains, in plain language and with almost no equations, every concept the later chapters rely on: black holes and horizons, curved spacetime, spin and the shadow, the accretion disk and its colours, stellar orbits, and the statistical toolkit. Each primer chapter ends with a short list of the terms just introduced and a pointer to the chapter where the full treatment lives. Readers comfortable with general relativity can skip Part I entirely.

Part II — Theory and Methods derives, from first principles, every formula the simulator implements: the Kerr geometry and its characteristic radii (Chapter 8), the Carter-separated geodesics and their second-order Mino-time form (Chapter 9), and the numerical integration machinery with its error analysis (Chapter 10). Each derivation cites the source function that implements it.

Part III — Results presents the science: shadow and critical-curve metrology (Chapter 11), the relativistic thin disk (Chapter 12), photon-ring substructure (Chapter 13), the S-star forward model (Chapter 14), the S2 fit and its statistics (Chapter 15), the wider S-star cluster (Chapter 16), and the no-hair constraint (Chapter 17).

Part IV — Engineering, validation, and closing documents the GPU implementation (Chapter 18), the Sgr A* calibration (Chapter 19), the interactive instrument (Chapter 20), the development history and complete test suite (Chapter 21), and closes with discussion and conclusions. The appendices carry extended algebra, algorithm pseudocode, number provenance, the reproduction pipeline, and a glossary.

A reader who wants only the headline science should read Chapters 11, 15 and 17, plus the calibration of Chapter 19. A reader evaluating the numerics should read Chapters 10 and 21. Every quantitative statement that is not a cited literature value is typeset from a generated macro and is reproducible by `make reproduce` (Appendix D).

Chapter 8

Kerr Geometry from First Principles

8.1 The Kerr solution in Boyer–Lindquist coordinates

In Boyer–Lindquist (BL) coordinates with $G = c = 1$ and spin parameter $a = J/M$, the Kerr line element is

$$\begin{aligned} \mathrm{d}s^2 = &-\left(1 - \frac{2Mr}{\Sigma}\right)\mathrm{d}t^2 - \frac{4Mar\sin^2\theta}{\Sigma}\,\mathrm{d}t\,\mathrm{d}\phi + \frac{\Sigma}{\Delta}\,\mathrm{d}r^2 \\ &+ \Sigma\,\mathrm{d}\theta^2 + \frac{A\sin^2\theta}{\Sigma}\,\mathrm{d}\phi^2, \end{aligned} \tag{8.1}$$

with $\Sigma = r^2 + a^2\cos^2\theta$, $\Delta = r^2 - 2Mr + a^2$ and $A = (r^2 + a^2)^2 - a^2\Delta\sin^2\theta$. This is the form implemented by `kerr_components` in `physics/src/metric.cpp`, and the determinant identity $\det g = -\Sigma^2\sin^2\theta$ is asserted by a dedicated unit test (Chapter 21).

> *[Chapter under construction — scope: Kerr–Schild form and the BL↔KS transformation; inverse metric; Killing vectors $\partial_t, \partial_\phi$; the Killing–Yano and Killing tensors as the origin of the Carter constant; frame dragging and the ZAMO angular velocity $\omega = -g_{t\phi}/g_{\phi\phi}$. See `paper/THESIS_PLAN.md`.]*

8.2 Horizons, ergosphere, and characteristic radii

The outer horizon lies at $r_+ = M + \sqrt{M^2 - a^2}$, the ergosphere at $r_E(\theta) = M + \sqrt{M^2 - a^2\cos^2\theta}$, and the equatorial photon orbits at $r_{\mathrm{ph}}^{\pm} = 2M\{1 + \cos[\tfrac{2}{3}\cos^{-1}(\mp a/M)]\}$ (prograde/retrograde). The innermost stable circular orbit follows the closed form of Bardeen et al. (1972),

$$\begin{aligned} Z_1 &= 1 + (1 - a_*^2)^{1/3}\left[(1 + a_*)^{1/3} + (1 - a_*)^{1/3}\right], \\ Z_2 &= \sqrt{3a_*^2 + Z_1^2}, \\ r_{\mathrm{ISCO}} &= M\left[3 + Z_2 \mp \sqrt{(3 - Z_1)(3 + Z_1 + 2Z_2)}\right], \end{aligned} \tag{8.2}$$

with the upper sign for prograde orbits. At $a_* \equiv a/M = 0$ these reduce to $2M$, $2M$, $3M$ and $6M$ exactly, which the test suite asserts to machine precision before any imaging is attempted (Chapter 21); for $a_* > 0$ it asserts the prograde/retrograde split (e.g. distinct photon-orbit radii at $a_* = 0.5$ and 0.94) and the monotonic approach $r_{\mathrm{ISCO}} \to r_+$ as $a_* \to 1$. These radii are implemented by `r_horizon`, `r_ergosphere`, `r_photon` and `r_isco` in `metric.cpp`.

[Chapter under construction — scope: Full algebra: equatorial photon orbits from the effective-potential cubic and its trigonometric solution; the BPT ISCO derivation from $\mathrm{d}^2V/\mathrm{d}r^2 = 0$ through Z_1, Z_2; the marginally bound radius r_{mb}. New figure: characteristic radii vs a_* (needs `physics_tables`). See `paper/THESIS_PLAN.md`.]

8.3 Circular-orbit energetics and accretion efficiency

[Chapter under construction — scope: $E(r)$, $L(r)$, $\Omega(r)$ for circular equatorial orbits (implemented by `energy_circular`, `angmom_circular`, `omega_circular`); binding energy at the ISCO and the accretion efficiency $\eta = 1 - E_{\mathrm{ISCO}}$ (5.7 per cent Schwarzschild → 42 per cent extremal). New figures: effective potential $V(r)$ at several spins; E, L, Ω, η vs r and vs a_*. See `paper/THESIS_PLAN.md`.]

Chapter 9

Geodesics and the Carter Constant

9.1 Constants of motion and the Carter separation

Geodesics in Kerr admit four constants of motion: the energy E, the axial angular momentum L_z, the Carter constant Q (Carter, 1968), and the norm μ^2 of the four-momentum. With $P(r) = E(r^2 + a^2) - aL_z$ the Carter-separated equations for a photon ($\mu = 0$) read

$$\begin{aligned}
\Sigma \frac{\mathrm{d}r}{\mathrm{d}\lambda} &= \pm\sqrt{R(r)}, \\
\Sigma \frac{\mathrm{d}\theta}{\mathrm{d}\lambda} &= \pm\sqrt{\Theta(\theta)}, \\
\Sigma \frac{\mathrm{d}\phi}{\mathrm{d}\lambda} &= -\Big(aE - \frac{L_z}{\sin^2\theta}\Big) + \frac{a}{\Delta}P(r), \\
\Sigma \frac{\mathrm{d}t}{\mathrm{d}\lambda} &= -a(aE\sin^2\theta - L_z) + \frac{(r^2+a^2)}{\Delta}P(r),
\end{aligned} \tag{9.1}$$

with the potentials

$$R(r) = P(r)^2 - \Delta\left[(L_z - aE)^2 + Q\right], \tag{9.2}$$

$$\Theta(\theta) = Q - \cos^2\theta\left[a^2(-E^2) + \frac{L_z^2}{\sin^2\theta}\right]. \tag{9.3}$$

The potentials and their derivatives are implemented by `radial_potential`, `polar_potential` and their `_deriv` companions in `physics/src/null_geodesics.cpp`.

> *[Chapter under construction — scope: The Hamilton–Jacobi derivation of the separation; Q from the Killing–Yano tensor; turning points and orbit classification by the roots of $R(r)$. New figures: sample photon and timelike trajectories; turning-point structure of $R(r)$. See `paper/THESIS_PLAN.md`.]*

9.2 Mino time and the second-order formulation

Rather than integrating the first-order radial and polar equations with explicit sign bookkeeping at turning points, we adopt Mino time τ (Mino, 2003), defined by $\mathrm{d}\lambda = \Sigma\,\mathrm{d}\tau$, which decouples the (r) and (θ) sectors, and evolve the *second-order* equations obtained by differentiating $\dot{r}^2 = R(r)$ once:

$$\frac{\mathrm{d}^2 r}{\mathrm{d}\tau^2} = \tfrac{1}{2}R'(r), \qquad \frac{\mathrm{d}^2\theta}{\mathrm{d}\tau^2} = \tfrac{1}{2}\Theta'(\theta), \tag{9.4}$$

in which turning points are ordinary smooth extrema and no square root of a near-zero quantity is ever taken. The first integrals $\dot{r}^2 = R(r)$ and $\dot{\theta}^2 = \Theta(\theta)$ then serve as exact algebraic constraints on the numerical solution, which we re-impose after every accepted step (Chapter 10).

The coordinate singularities at $\theta = 0, \pi$ are regularised with a 10^{-8} clamp; they are properties of the BL chart, not of the spacetime.

9.3 Timelike geodesics

For timelike geodesics ($\mu = 1$, stellar orbits) the radial potential gains the mass term, $R(r) = P^2 - \Delta[r^2 + (L_z - aE)^2 + Q]$, the polar potential becomes $\Theta = Q - \cos^2\theta[a^2(1-E^2) + L_z^2/\sin^2\theta]$, and the four-velocity normalisation $g_{\mu\nu}u^\mu u^\nu = -1$ provides an independent per-step health check (`solve_ut` in `timelike_geodesics.cpp` solves the normalisation quadratic; `carter_q` recovers Q from the state). We integrate stellar orbits in the full geodesic equations rather than any post-Newtonian expansion: at the S2 pericentre ($\sim$1470 Schwarzschild radii, orbital speed 2.6 per cent of c) the leading PN corrections are individually at the 10^{-4} level — the same order as the measured redshift excess itself — so the expansion error budget would have to be argued term by term, whereas the geodesic integrator's error budget is simply its conservation diagnostics.

> *[Chapter under construction — scope: [OPTIONAL] The exact closed-form solution in elliptic integrals / Weierstrass $\wp$ (Rauch–Blandford, Gralla–Lupsasca) as an analytic integrator check; Mino-time vs affine-parameter figure. See `paper/THESIS_PLAN.md`.]*

Chapter 10

Numerical Methods and Error Analysis

10.1 The Dormand–Prince 5(4) integrator

All geodesics are integrated with an adaptive Dormand–Prince 5(4) Runge–Kutta scheme (`DormandPrince54::try_step` in `physics/include/sgr_physics/integrator.hpp`): the embedded fourth-order solution provides the local error estimate, and a PI controller adapts the step to hold the mixed error norm $\epsilon = \max_i |y_i^{(5)} - y_i^{(4)}|/(\mathrm{atol} + \mathrm{rtol}\,|y_i|)$ at unity. For null rays used in metrology the tolerances are 10^{-11} (absolute and relative).

[Chapter under construction — scope: The full Butcher tableau; derivation of the embedded fourth-order error estimate; PI step-control stability. New figures (need `convergence_tool`): conserved-quantity drift vs step size (log–log); step-size adaptation $h(\lambda)$ along a ray. See `paper/THESIS_PLAN.md`.]

10.2 Constraint projection

The decisive ingredient for shadow work is *constraint projection*: after every accepted step the radial and polar Mino-time momenta are rescaled to satisfy the first integrals exactly,

$$\dot{r} \;\leftarrow\; \mathrm{sgn}(\dot{r})\,\sqrt{R(r)}, \qquad \dot{\theta} \;\leftarrow\; \mathrm{sgn}(\dot{\theta})\,\sqrt{\Theta(\theta)}, \tag{10.1}$$

preserving the evolved sign (and leaving the momentum untouched within a turning-point guard band where R or Θ dips below zero by less than the local error). Without projection, the small $\mathcal{O}(\mathrm{rtol}\cdot r^4)$ drift accumulated crossing $\sim 10^3\,M$ of nearly flat spacetime between the camera (at $r = 1000\,M$) and the hole systematically mis-classifies rays near the critical impact parameter — in a controlled experiment the capture/escape classification of boundary rays flipped wholesale as the tolerance was varied. With projection the measured boundary is independent of the integrator tolerance. Conservation of (E, L_z, Q) is asserted to 10^{-6} relative along every test geodesic, and $|g_{\mu\nu}u^\mu u^\nu + 1| < 10^{-8}$ along timelike ones (both are hard gates in the test suite, Chapter 21).

[Chapter under construction — scope: Projection error analysis; the tolerance-dependence experiment as a figure: projection on/off shadow-boundary position vs tolerance (needs `convergence_tool`). See `paper/THESIS_PLAN.md`.]

10.3 Complex-step differentiation

Christoffel symbols for arbitrary metrics (needed for the Hartle–Thorne integrator) are evaluated by complex-step differentiation with step $h = 10^{-100}$ (`christoffel` in `metric.cpp`), which is exact to machine precision and free of subtractive cancellation.

> *[Chapter under construction — scope: The Taylor-expansion proof of machine-precision exactness (Squire–Trapp); comparison against central differences; convergence and Richardson studies. See `paper/THESIS_PLAN.md`.]*

Part III

Results

Chapter 11

The Reference Ray Tracer, Shadow, and Critical Curve

11.1 The ZAMO camera and ray launching

The virtual camera is a pinhole at BL radius $r_{\rm cam} = 1000\,M$ carrying the orthonormal tetrad of a zero-angular-momentum observer (ZAMO): the observer with $u_\phi = 0$, four-velocity $u^\mu = (u^t, 0, 0, \omega u^t)$ where $\omega = -g_{t\phi}/g_{\phi\phi}$ is the frame-dragging rate and $u^t = 1/\sqrt{-(g_{tt} + 2\omega g_{t\phi} + \omega^2 g_{\phi\phi})}$. This tetrad reduces to the static observer at $a = 0$ and remains well defined inside the ergosphere (`zamo_tetrad` in `physics/src/raytrace_cpu.cpp`). A local photon direction $(n^{\hat r}, n^{\hat\theta}, n^{\hat\phi})$ on the unit sphere is contracted with the tetrad legs to give the BL four-momentum, from which (E, L_z, Q) follow algebraically (`camera_ray`); rays are launched forward (null geodesics are reversible, so where a ray lands is what the pixel sees).

11.2 Grid-free boundary metrology

The shadow boundary is measured without any pixel grid: along each image-plane direction ψ the captured/escaped transition is bracketed and bisected in camera angle, giving the critical angle $\alpha(\psi)$ to $\sim 10^{-9}$ rad (`shadow_boundary_angle`). Angular radii are converted to impact parameters via $b = r\sin\alpha/\sqrt{1 - 2M/r}$ — the flat-space relation corrected for the gravitational bending of the reference 'straight line' at finite camera radius — and quoted as diameters in units of M; the conversion to sky angle uses the GRAVITY mass and distance (Chapter 19).

11.3 Schwarzschild validation: the Synge formula

For $a_* = 0$ the critical impact parameter follows in three lines: the photon sphere sits where the effective potential $V(r) = (1 - 2M/r)/r^2$ peaks, $V'(r) = 0 \Rightarrow r_{\rm ph} = 3M$; the impact parameter of a circular photon orbit there is $b = 1/\sqrt{V(r_{\rm ph})} = 3M/\sqrt{1/3} = \sqrt{27}\,M$; and for an observer at finite radius the shadow angular radius is $\sin\alpha_{\rm sh} = \sqrt{27}\,M\sqrt{1 - 2M/r_{\rm obs}}/r_{\rm obs}$ (Synge's formula; see also Luminet, 1979) (`schwarzschild_shadow_angle`). The traced boundary agrees with this to a relative deviation of 5×10^{-16} — machine precision — and is circular to 0.000 per cent over 32 directions, with an equivalent diameter of $10.3924\,M$ (the analytic value is $2\sqrt{27}\,M \simeq 10.3923\,M$). The same $2\sqrt{27}\,M$ anchor reappears independently in the (a_*, i) sweep of Chapter 19: every $a_* = 0$ grid row returns it identically at all inclinations, as it must by spherical symmetry.

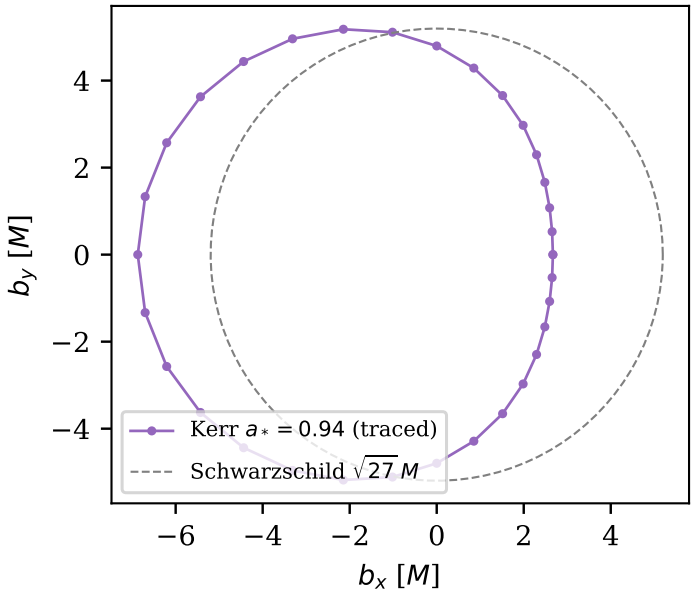


Figure 11.1: Traced Kerr shadow boundary at $a_* = 0.94$, edge-on, against the analytic Bardeen (1973) critical curve. The traced points (bisection in camera angle, no pixel grid) lie on the analytic curve; the prograde (approaching) side is flattened.

11.4 The Kerr shadow and the Bardeen critical curve

At $a_* = 0.94$ the shadow is displaced and flattened on the prograde side, in quantitative agreement with the analytic critical curve of Bardeen (1973): the traced boundary reproduces $\xi(r) = [r^2(3M - r) - a^2(r + M)]/[a(r - M)]$, $\eta(r) = r^3[4a^2M - r(r - 3M)^2]/[a^2(r - M)^2]$ point-by-point (Fig. 11.1). Viewed edge-on the mean diameter is $9.5445\,M$.

[Chapter under construction — scope: Derivation of the Bardeen $\xi(r), \eta(r)$ curve from the double-root condition on $R(r)$; critical-curve overlays for $a_* = 0, 0.5, 0.94, 0.998$ (needs the `sweep_tool` extension). See `paper/THESIS_PLAN.md`.]

11.5 Spherical photon orbits and Lyapunov exponents

[Chapter under construction — scope: The full family of unstable spherical photon orbits at all inclinations; the Lyapunov exponent γ of bound photon orbits and the sub-ring demagnification $e^{-\gamma}$. New figure: spherical photon-orbit radii and γ vs a_*. See `paper/THESIS_PLAN.md`.]

11.6 Gravitational lensing of the disk and sky

[Chapter under construction — scope: User-requested topic (THESIS_PLAN Part 1c): why the far side of the disk appears above and below the shadow; the Einstein-ring limit; the ray-bending schematic (from `make_diagrams.py`) plus the traced lensed-disk render. See `paper/THESIS_PLAN.md`.]

Chapter 12

The Accretion Disk: Novikov–Thorne and Page–Thorne

12.1 The Novikov–Thorne disk and the Page–Thorne flux

The accretion flow is modelled as the geometrically thin, optically thick equatorial disk of Novikov & Thorne (1973), using the closed-form time-averaged flux profile of Page & Thorne (1974), their eq. (15n): in terms of $x = \sqrt{r/M}$ and the roots $x_{1,2,3}$ of $x^3 - 3x + 2a_* = 0$,

$$F(r) \propto \frac{1}{x^4\,(x^3 - 3x + 2a_*)}\Big[x - x_0 - \tfrac{3}{2}a_* \ln\frac{x}{x_0} - \sum_{i=1}^{3} \frac{3(x_i - a_*)^2}{x_i(x_i - x_j)(x_i - x_k)} \ln\frac{x - x_i}{x_0 - x_i}\Big], \tag{12.1}$$

with $x_0 = \sqrt{r_{\rm ISCO}/M}$ (implemented by `flux` in `physics/src/disk_model.cpp`). Two numerical notes: the root $x_2 \to 0$ as $a_* \to 0$ and its logarithm coefficient must be guarded at the Schwarzschild limit; and the far-field expansion is *not* the Newtonian $1 - \sqrt{r_{\rm in}/r}$ profile — the GR log terms shift the constant, $f(x) \to x - c + 3/x + \mathcal{O}(x^{-2})$ with $c \simeq 3.976$ for $a_* = 0$, which the unit tests assert against the exact expansion rather than the Newtonian folklore form. The profile is evaluated between $r_{\rm ISCO}$ and an outer radius $r_{\rm out}$.

> *[Chapter under construction — scope: The thin-disk conservation laws (mass, energy, angular momentum) leading to the radial structure; the full $f(x)$ derivation with the x_1, x_2, x_3 roots and log terms. New figures: $F(r)$ and $T(r)$ profiles at several a_* (needs `physics_tables`). See `paper/THESIS_PLAN.md`.]*

12.2 Redshift factor and Doppler beaming

Each backward-traced ray that crosses the equatorial annulus is assigned the redshift factor

$$g \equiv \frac{\nu_{\rm obs}}{\nu_{\rm emit}} = \frac{\sqrt{-(g_{tt} + 2\Omega g_{t\phi} + \Omega^2 g_{\phi\phi})}}{1 - \Omega\, b_\gamma}, \tag{12.2}$$

where Ω is the circular-orbit angular velocity at the crossing radius and $b_\gamma = L_z/E$ the photon's impact parameter. Since a blackbody redshifted by g is exactly a blackbody at temperature gT, the observed bolometric intensity scales as g^4 (Doppler beaming, `bolometric` in `disk_model.cpp`), and the colour follows from the Planck spectrum at $g\,T_{\rm eff}(r)$ through the CIE 1931 observer (`blackbody_rgb`). Photon sub-rings are labelled by the number n of equatorial crossings prior to the terminating one (Johnson et al., 2020); their metrology is the subject of Chapter 13.

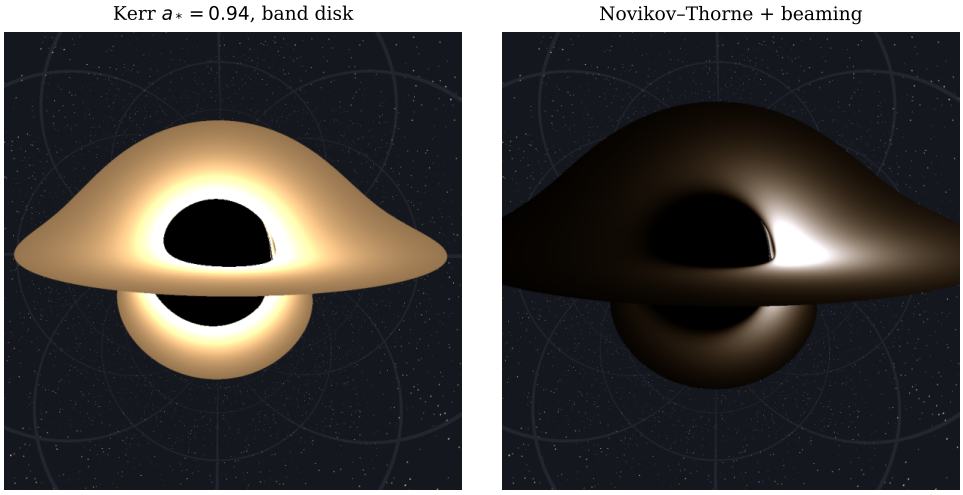


Figure 12.1: CPU reference render of the Novikov–Thorne disk at $a_* = 0.94$, $i = 80°$ (linear sRGB from Planck spectra at $g\,T_{\rm eff}$). Doppler beaming brightens the approaching limb by a bolometric factor of 9.60.

[Chapter under construction — scope: Derivation of the g^4 law from Liouville invariance of I_ν/ν^3; the Planck-to-CIE colour pipeline. New figures: g-factor image map; blackbody/CIE locus. See `paper/THESIS_PLAN.md`.]

12.3 The rendered disk

Fig. 12.1 shows the Novikov–Thorne disk at $a_* = 0.94$, $i = 80°$. The redshift factor spans $g = 0.139$ to 1.440 across the image, and the resulting g^4 beaming makes the approaching side 9.60× brighter in bolometric flux than the receding side — the classic one-sided appearance anticipated by Luminet (1979) and rendered cinematically by James et al. (2015).

12.4 Spin morphology of the disk image

[Chapter under construction — scope: User-requested topic (THESIS_PLAN Part 1c): at $a_* = 0$ the wound streamlines of the display layer form smooth, continuous, near-concentric lines around the whole disk; at high spin ($a_* \approx 0.906$) the coherent bright streamlines collapse to a small patch on the approaching side — the ISCO moves inward, frame dragging and g^4 beaming intensify and asymmetrise. Awaiting the user's two comparison engine renders; captions will carry the exact `sgr_engine` command line that regenerates each. See `paper/THESIS_PLAN.md`.]

Chapter 13

Photon-Ring Substructure

13.1 Sub-ring decomposition by crossing count

Binning image pixels by the number of prior equatorial crossings isolates the photon sub-rings (Johnson et al., 2020): the direct image ($n = 0$), the first lensed ring ($n = 1$), and the second ring ($n = 2$) (`SubRingExtractor::analyze` in `science/photon_ring/`). The bolometric flux ratios are $F_0/F_1 = 10.7$ and $F_1/F_2 = 274$ (Fig. 13.1), the steep $n = 1 \to n = 2$ demagnification reflecting the exponential suppression $e^{-\gamma}$ per half-orbit near the critical curve, with γ the Lyapunov exponent of the bound photon orbits. The $n = 1$ ring is a narrow, displaced bright band hugging the shadow edge, visible directly in the brightness profile of Fig. 13.2; its measured radial width at this geometry is a few per cent of the shadow diameter (`scan_widths`), qualitatively the regime in which Johnson et al. (2020) project that space–ground interferometry could detect the $n = 1$ signature in the visibility-domain power-law tail. Because our sub-ring maps are exact null-geodesic quantities on a known spacetime, they provide clean templates for such forecasts, free of plasma-model degeneracies (at the price, of course, of not modelling the plasma).

13.2 Sub-ring metrics across the spin range

[Chapter under construction — scope: Sub-ring flux ratio and width vs a_* (needs the `subring_tool --sweep` extension); the Johnson et al. universal parameters; connection to the Lyapunov exponents of Chapter 11. See `paper/THESIS_PLAN.md`.]

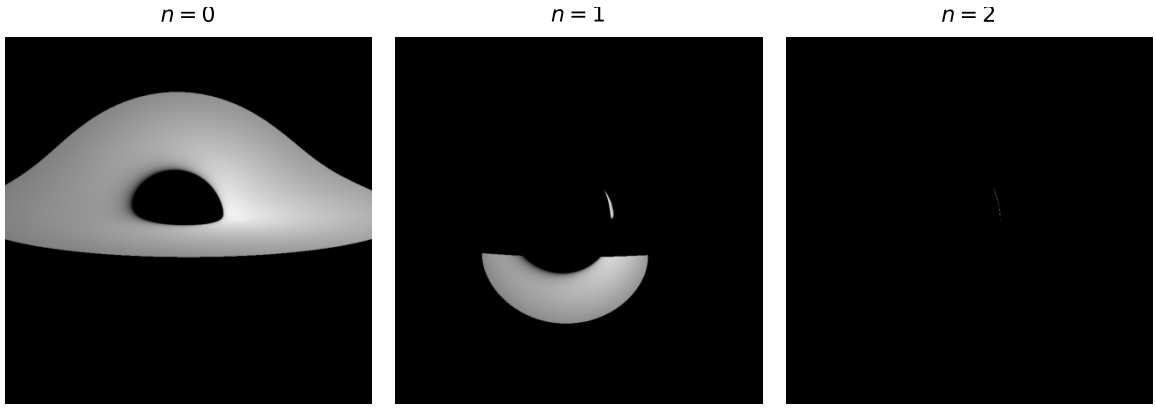


Figure 13.1: Bolometric maps of the photon sub-rings $n = 0, 1, 2$ (log stretch). Flux ratios: $F_0/F_1 = 10.7$, $F_1/F_2 = 274$.

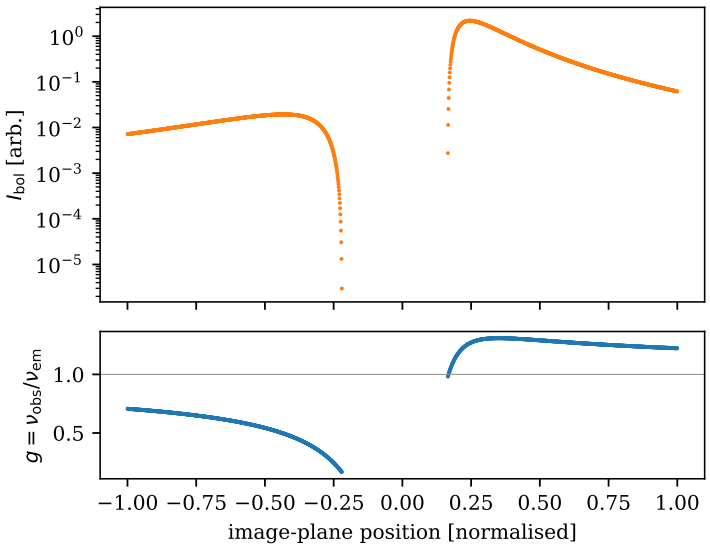


Figure 13.2: Bolometric intensity along the image major axis. The narrow $n = 1$ sub-ring stands on the shoulder of the direct disk image at the shadow edge.

13.3 The axial image tower

[Chapter under construction — scope: User-requested topic (THESIS_PLAN Part 1c): the bright vertical columns along the spin axis seen in near-edge-on renders are the gravitationally-lensed polar caustic — stacked higher-order images of the far/underside of the disk funnelled along the axis — NOT an astrophysical jet. Contrast with real relativistic jets (Blandford–Znajek, MHD-driven), which require plasma physics outside this thin-disk + geodesic model. Awaiting the user's engine render; the caption will carry the exact `sgr_engine` *reproduction command. See* `paper/THESIS_PLAN.md`*.]*

13.4 Detectability forecast

[Chapter under construction — scope: ngEHT-class detectability of the $n = 1$ signature; visibility-domain power-law tail; what the exact-geodesic templates provide. See `paper/THESIS_PLAN.md`*.]*

Chapter 14

Timelike Orbits and the S-Star Forward Model

14.1 The forward model

The S2 model integrates the timelike geodesic of Chapter 9 in the fitted spacetime, projects it on the sky (east–north offsets relative to the mass), and predicts radial velocities including the special-relativistic transverse Doppler shift and the gravitational redshift (`SkyProjection::S2Model` in `science/orbit_fitter/`).

14.2 Orbital elements from radial turning points

Orbital elements are converted to initial conditions through the exact radial turning points: E and L_z follow from the two conditions $R(r_{\rm peri}) = R(r_{\rm apo}) = 0$ (Darwin-style elements), a 2×2 linear system in (E^2, L_z^2) for equatorial motion. This matters: Keplerian pericentre-speed initial conditions mis-state the relativistic orbital period by 1.6 per cent at the S2 eccentricity — two orders of magnitude above the measurement precision of the period (the bug-hunt narrative is told in Chapter 21).

> *[Chapter under construction — scope: Full derivation of the (E^2, L_z^2) linear system from the turning-point conditions; Hermite interpolation of the sampled geodesic (`sample_at`). See `paper/THESIS_PLAN.md`.]*

14.3 Sky projection and the Rømer delay

Light travel time across the orbit (the Rømer delay, $\Delta t = -z \sin i / c$, $\sim$8 light-days peak-to-peak for S2) is solved per epoch by fixed-point iteration on the emission time $t_{\rm em} = t_{\rm obs} - \Delta t(t_{\rm em})$, which converges in 2–3 iterations at S2 velocities.

> *[Chapter under construction — scope: Derivation of the delay from the line-of-sight displacement; convergence of the fixed-point iteration. New figure: Rømer delay curve over one S2 orbit. See `paper/THESIS_PLAN.md`.]*

14.4 The spectroscopic observable

The measured quantity for S2 spectroscopy is the centroid of the Brγ line ($\lambda_{\rm rest} = 2.16612\,\mu$m). The model wavelength ratio factorises as

$$\frac{\lambda_{\rm obs}}{\lambda_{\rm emit}} = (1 + z_{\rm D})(1 + z_{\rm tD})(1 + z_{\rm grav}), \tag{14.1}$$

where z_{D} is the classical line-of-sight Doppler term, z_{tD} the special-relativistic transverse (time-dilation) term and z_{grav} the gravitational redshift at the emission point (`redshift_terms`; the line model is `SpectralLineFitter::predict`). The quoted model radial velocity is the apparent $v_r = c\,(\lambda_{\mathrm{obs}}/\lambda_{\mathrm{rest}} - 1)$, the same convention the observers use, so the relativistic terms appear as an excess over a Keplerian orbit with identical elements (Section 15.4).

[Chapter under construction — scope: New figure: the redshift decomposition (gravitational vs transverse vs classical) stacked over one orbit — the artifact `phase4/s2_redshift_curve.csv` already carries the columns. See `paper/THESIS_PLAN.md`.]

Chapter 15

Orbit Fitting, Statistics, and the S2 Results

15.1 Data

The astrometric and spectroscopic record of S2 is taken from the monitoring compilation of Gillessen et al. (2017) (VizieR J/ApJ/837/30): 145 imaging positions (VLT NACO and Keck NIRC, 1992–2016, typical uncertainties 0.3–3 mas) and 44 radial velocities (SINFONI and earlier spectrographs, 2000–2016, typical uncertainties 15–100 $\mathrm{km\,s^{-1}}$). Positions are offsets from the radio position of Sgr A*; the residual uncertainty of that frame tie is modelled by the fitted offset/drift parameters with the Plewa et al. (2015) priors of Section 15.2. We apply no error rescaling: the reduced χ^2 of 0.959 below is obtained with the published uncertainties as-is.

15.2 The objective and its parameterisation

The objective is

$$\chi^2 = \sum_i \left[\frac{(\alpha_i^{\mathrm{obs}} - \alpha_i^{\mathrm{mod}})^2}{\sigma_{\alpha,i}^2} + \frac{(\delta_i^{\mathrm{obs}} - \delta_i^{\mathrm{mod}})^2}{\sigma_{\delta,i}^2} \right] + \sum_j \frac{(v_{r,j}^{\mathrm{obs}} - v_{r,j}^{\mathrm{mod}})^2}{\sigma_{v,j}^2} + \chi^2_{\mathrm{prior}}, \tag{15.1}$$

over the 145 astrometric epochs and 44 radial velocities, with the forward model of Chapter 14 supplying the model positions and velocities.

Thirteen parameters are fitted: (M, R_0), six orbital elements, the four reference-frame offsets and drifts (x_0, y_0, v_x, v_y) of the mass with respect to the radio frame, and v_z; the no-hair mode adds ε_q. Weak priors tie the frame to Sgr A* following Plewa et al. (2015) (± 0.2 mas at epoch 2009.0, ± 0.1 $\mathrm{mas\,yr^{-1}}$, ± 5 $\mathrm{km\,s^{-1}}$). The χ^2 is minimised with the Ceres solver (automatic differentiation; `OrbitFitter::fit`); parameter covariances are obtained as $C = (J^{\mathsf{T}} J)^{-1}$ from an explicit central-difference Jacobian J of the weighted residual vector (`OrbitFitter::covariance`), because the solver's truncated-SVD covariance annihilates the strongly correlated M–R_0 direction and reports a spuriously zero mass uncertainty (Chapter 21).

[Chapter under construction — scope: Fisher information and the Cramér–Rao bound; why SVD truncation kills the M–R_0 direction, worked in the singular basis. See `paper/THESIS_PLAN.md`.]

Table 15.1: S2 fit results (full GR timelike geodesic model) against published values (Gillessen et al., 2017; GRAVITY Collaboration, 2022). Uncertainties are 1σ from the explicit-Jacobian covariance.

Parameter	This work	Published
M [$10^6\,\mathrm{M}_\odot$]	4.270 ± 0.195	4.297 ± 0.012
R_0 [pc]	8272 ± 182	8277 ± 9
a [mas]	124.82 ± 0.88	125.058 ± 0.041
e	0.88277 ± 0.00195	0.8839 ± 0.0019
i [deg]	134.70 ± 0.41	134.18 ± 0.40
Ω [deg]	227.33 ± 0.63	226.94 ± 0.60
ω [deg]	65.44 ± 0.61	65.51 ± 0.57
t_{peri} [yr]	2002.3227 ± 0.0052	2002.33 ± 0.01
P [yr]	16.0561 ± 0.0228	16.0518 ± 0.0022
χ^2_ν	0.959	—

15.3 Results: the S2 fit

We fit the 145 astrometric positions and 44 radial velocities of S2 tabulated by Gillessen et al. (2017). The best-fitting model (Fig. 15.1, Fig. 15.2) has $\chi^2 = 307.8$ for 321 degrees of freedom ($\chi^2_\nu = 0.959$); the residuals (Fig. 15.3) show no structure. Table 15.1 compares the recovered parameters with the published values.

The recovered mass and distance agree with the far more precise GRAVITY interferometric determination (GRAVITY Collaboration, 2022) to 0.14σ and 0.03σ respectively — a non-trivial cross-check, since our fit uses none of the GRAVITY beam astrometry, only the VLT/Keck imaging era compiled by Gillessen et al. (2017).

> *[Chapter under construction — scope: New figures: sky-plane residual quiver; observed-minus-computed vs time. See `paper/THESIS_PLAN.md`.]*

15.4 Gravitational redshift at pericentre

The magnitude of the relativistic signal at pericentre follows from the fitted elements alone. Step by step:

1. The pericentre distance is $r_{\mathrm{peri}} = a(1-e) = 1032\,\mathrm{au} \times (1 - 0.88277) \approx 121$ au, i.e. ≈ 1470 Schwarzschild radii ($R_S = 2GM/c^2 \approx 0.085$ au).
2. The vis-viva speed there is $v_{\mathrm{peri}} = \sqrt{GM(2/r_{\mathrm{peri}} - 1/a)} \approx 7650\,\mathrm{km\,s^{-1}}$, so $\beta \approx 0.0255$.
3. The transverse Doppler shift contributes $z_{\mathrm{tD}} \simeq \beta^2/2 \approx 3.3 \times 10^{-4}$, i.e. $c\,z_{\mathrm{tD}} \approx 98\,\mathrm{km\,s^{-1}}$.
4. The gravitational redshift contributes $z_{\mathrm{grav}} \simeq GM/(r_{\mathrm{peri}}c^2) = \beta^2_{\mathrm{circ}} \approx 3.5 \times 10^{-4}$, i.e. $\approx 105\,\mathrm{km\,s^{-1}}$.

The two corrections are nearly equal at pericentre — not a coincidence: vis-viva gives $v^2_{\mathrm{peri}} = (GM/a)(1+e)/(1-e)$ while $GM/r_{\mathrm{peri}} = (GM/a)/(1-e)$, so $z_{\mathrm{grav}}/z_{\mathrm{tD}} = 2/(1+e) = 1.06$ at the S2 eccentricity — and their sum in the full geodesic model peaks at $205\,\mathrm{km\,s^{-1}}$ at epoch 2018.379 (Fig. 15.4), in agreement with the $\approx 200\,\mathrm{km\,s^{-1}}$ excess measured around the 2018.38 pericentre passage by GRAVITY Collaboration (2018), who quantified it as $f = 0.90 \pm 0.17$ on a scale where $f = 0$ is Newtonian and $f = 1$ is GR. Our forward model contains no adjustable redshift amplitude: the excess is forced by the fitted mass and geometry. (Like GRAVITY Collaboration (2018) we neglect the $\sim$5 $\mathrm{km\,s^{-1}}$ light-bending correction.)

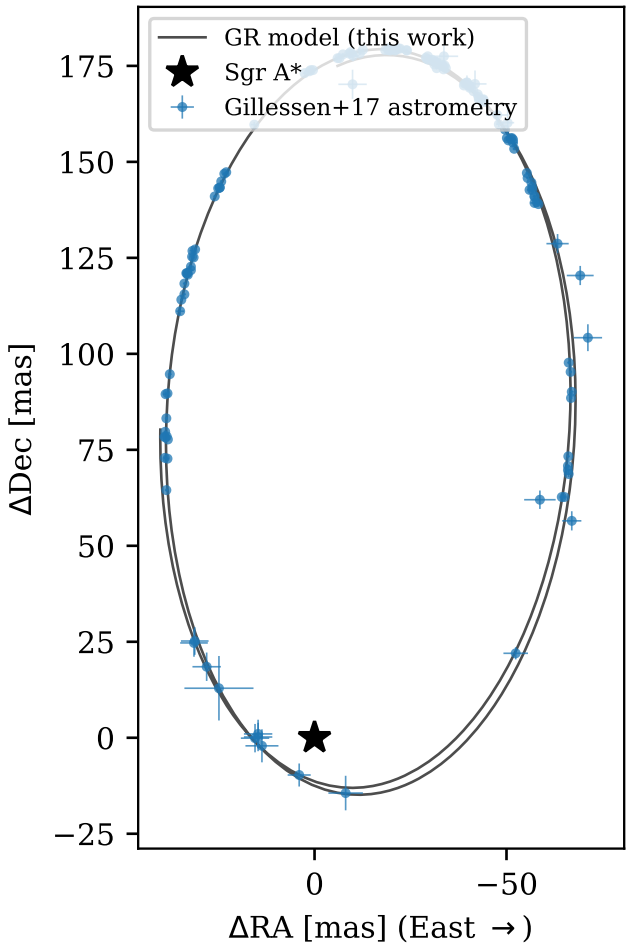


Figure 15.1: S2 on the sky: fitted GR geodesic model (curve) and the 145 astrometric measurements of Gillessen et al. (2017) (points). Sgr A* sits at the origin.

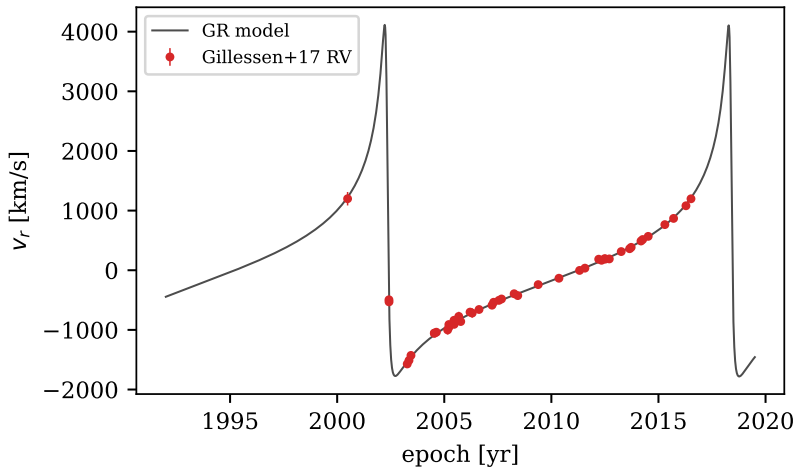


Figure 15.2: S2 radial velocity: model and the 44 measurements. The sharp swing brackets the 2002 and 2018 pericentre passages.

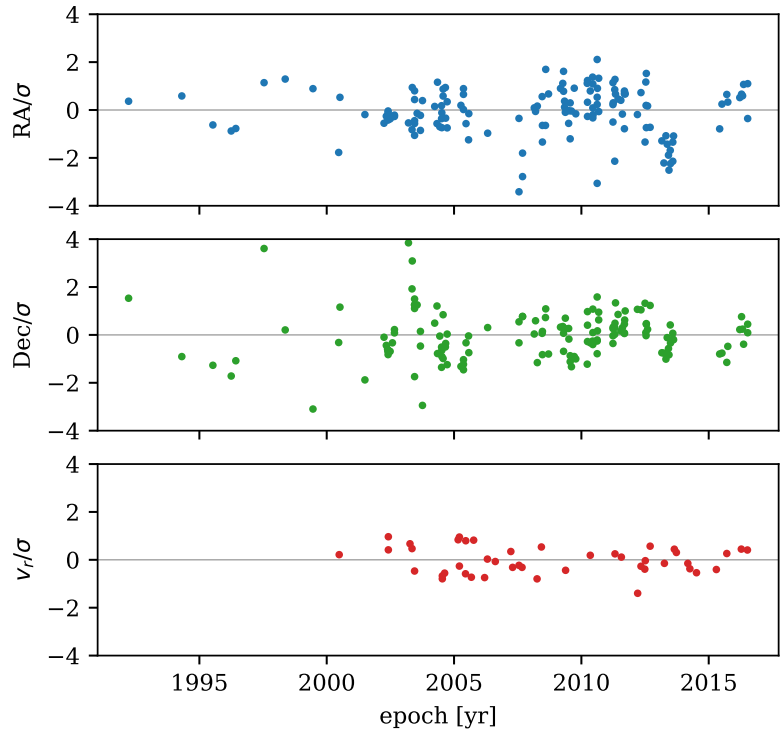


Figure 15.3: Normalised fit residuals. $\chi^2_\nu = 0.959$ over 321 degrees of freedom.

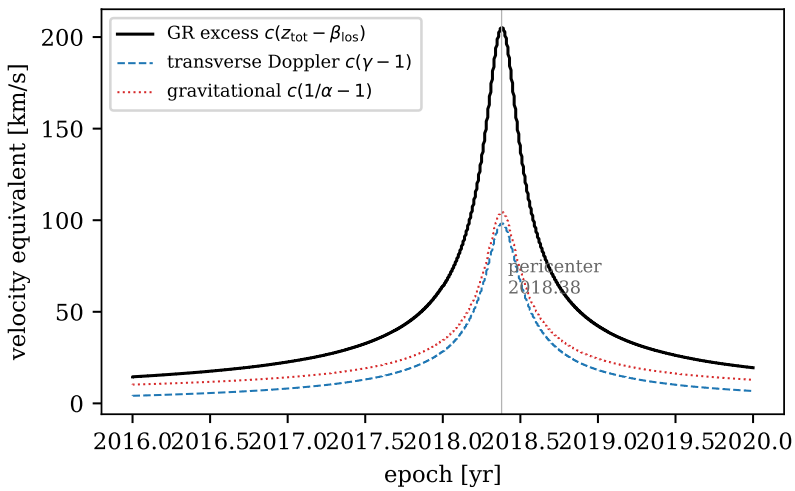


Figure 15.4: Relativistic radial-velocity excess of S2 over a Keplerian orbit with identical elements, through the 2018 pericentre passage. The peak is $205\,\mathrm{km\,s^{-1}}$ at epoch 2018.379, matching the detection of GRAVITY Collaboration (2018).

15.5 Relativistic precession

The fitted orbit advances its pericentre by 11.98 arcmin per orbital period from the Schwarzschild term. The textbook estimate agrees step by step: the semilatus rectum is $p = a(1 - e^2) = 1032\,\mathrm{au} \times (1 - 0.88277^2) \approx 228$ au, so with $R_g \approx 0.0424$ au the ratio is $R_g/p = 1.86 \times 10^{-4}$ and

$$\delta\omega = \frac{6\pi GM}{c^2 a(1-e^2)} = 6\pi \frac{R_g}{p} \approx 3.5 \times 10^{-3}\,\mathrm{rad}, \tag{15.2}$$

i.e. ≈ 12.0 arcmin per 16-yr orbit, matching the full geodesic integration (11.98 arcmin) and the GRAVITY detection of the Schwarzschild precession, $f_{\mathrm{SP}} = 1.10 \pm 0.19$ (GRAVITY Collaboration, 2020). Frame dragging at $a_* = 0.94$ contributes only -0.204 arcmin per orbit (retrograde for the S2 geometry), two orders of magnitude below current sensitivity, consistent with the hierarchy laid out by Will (2008).

[Chapter under construction — scope: New figure: the precession decomposition (Schwarzschild, Lense–Thirring, quadrupole) from `GRPrecession`'s three analytic terms — a pure formula evaluation, no new tool. See `paper/THESIS_PLAN.md`.]

15.6 MCMC posteriors

[Chapter under construction — scope: [CORE, THESIS_PLAN Part 2] Affine-invariant ensemble sampling (Goodman–Weare / emcee) of the 13-parameter posterior via a `--chi2` evaluation mode on `fit_s2_tool`; autocorrelation and convergence diagnostics; posterior vs the Jacobian covariance. New figure: the S2 corner plot. See `paper/THESIS_PLAN.md`.]

Chapter 16

The Wider S-Star Cluster: A Joint Mass Fit

[Chapter under construction — scope: [CORE, THESIS_PLAN Part 2] Extend the data pipeline to further S-stars from the VizieR J/ApJ/837/30 tables (flag-parse refactor of `fetch_gillessen_data.py`); the shared-potential model (one M, R_0, per-star orbital elements); the joint likelihood (`fit_cluster` tool); the combined mass posterior against the S2-only result and the GRAVITY 2022 multi-orbit value. Any use of S62 will be labelled contested. New figures: all fitted S-star orbits on the sky; per-star and joint mass posteriors; mass vs number-of-stars convergence. See `paper/THESIS_PLAN.md`.]

Chapter 17

A No-Hair Test with a Free Quadrupole

17.1 The Hartle–Thorne / quasi-Kerr metric

To parametrise departures from Kerr we use the quasi-Kerr form of the Hartle–Thorne metric (Hartle & Thorne, 1968; Glampedakis & Babak, 2006): the total quadrupole is written

$$Q_{\rm tot} = -Ma^2 + \varepsilon_q M^3, \tag{17.1}$$

and the deviation enters as a perturbation h^{ab} of the *inverse* metric, valid through $\mathcal{O}(a^2, \varepsilon_q)$. At $\varepsilon_q = 0$ the implementation reproduces the Kerr tensor to 10^{-12} (a dedicated unit test), and for $\varepsilon_q \neq 0$ it produces the expected additional pericentre advance $\delta\omega_Q = -3\pi\varepsilon_q\,(M/p)^2$ per orbit for equatorial orbits of semilatus rectum p (Will, 2008), which anchors the sign convention of the perturbation. Because the far-field expansion of the two structure functions F_1, F_2 of Glampedakis & Babak (2006) suffers catastrophic cancellation at large r, the implementation (`physics/src/hartle_thorne.cpp`) switches to an asymptotic series beyond $r = 9M$.

> *[Chapter under construction — scope: Geroch–Hansen multipole moments and the no-hair statement; the Hartle–Thorne slow-rotation metric to $\mathcal{O}(a^2)$; the F_1, F_2 structure functions and their large-r cancellation, with the branch-continuity test at the $9M$ switch. See `paper/THESIS_PLAN.md`.]*

17.2 Profile-likelihood constraint from S2

Re-fitting the S2 data in the free-quadrupole Hartle–Thorne spacetime with ε_q as a fourteenth parameter uses a profile likelihood: ε_q is stepped over a grid, the remaining thirteen parameters are re-optimised at each step (`OrbitFitter::fit_nohair`), and $\Delta\chi^2(\varepsilon_q) = \chi^2(\varepsilon_q) - \chi^2_{\rm min}$ is interpolated to the $\Delta\chi^2 = 1$ (68 per cent) and 3.84 (95 per cent) crossings. The profile is flat at its Kerr value near $\varepsilon_q = 0$ (Fig. 17.1): the best fit is $\varepsilon_q = 0.00 \times 10^5$ with $\sigma(\varepsilon_q) = 0.93 \times 10^5$ and a 95 per cent confidence half-width of 1.33×10^5. The data are fully consistent with the no-hair value $\varepsilon_q = 0$, and the marginal χ^2_ν is unchanged from the Kerr fit — the extra degree of freedom buys no improvement, as expected if Kerr is the true spacetime.

17.3 Why the constraint is weak, and how it improves

The constraint is weak in absolute terms because the quadrupole pericentre shift scales as $\delta\omega_Q = -3\pi\,\varepsilon_q\,(M/p)^2$ per orbit (Will, 2008). Working the numbers: $M/p = R_g/p = 0.0424/228 = 1.86 \times 10^{-4}$, so $(M/p)^2 = 3.5 \times 10^{-8}$ and one unit of ε_q moves the pericentre by only $3.3 \times$

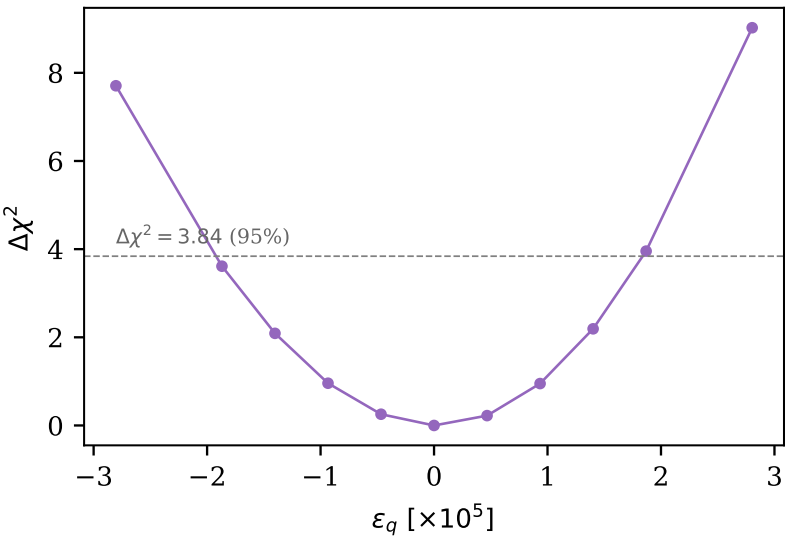


Figure 17.1: $\Delta\chi^2$ profile of the quadrupole deviation ε_q from the S2 data. The 95 per cent interval is $|\varepsilon_q| < 1.33 \times 10^5$, consistent with the Kerr value $\varepsilon_q = 0$.

10^{-7} rad per orbit — 0.07 arcsec, versus an astrometric sensitivity to ω of order 0.5° per orbit (Table 15.1). Inverting, the expected 1σ reach is $\sigma(\varepsilon_q) \sim (0.5^\circ)/(0.07'') \sim 10^5$, exactly the order recovered by the profile likelihood: a single star at ∼3000 gravitational radii is simply far from the quadrupole's domain of influence. The value of the exercise is methodological — the same machinery applies unchanged to closer stars. Continued monitoring helps too: propagating the fitted covariance forward, ten further S2 orbits (astrometry and RV of the quality of the existing series) shrink the uncertainty to $\sigma(\varepsilon_q) \approx 0.34 \times 10^5$, a factor ∼3. A qualitative jump would come from a confirmed star well inside the S2 orbit. The reported candidate S62, with a claimed 9.9-yr period and ∼36 au pericentre, would be such a star, but we emphasise: *S62 is a contested orbit — its identification is unconfirmed and may be a misidentification of S29 or a blend artefact*; we therefore do not build quantitative forecasts on it, and any future use of S62 in this context must carry the same caveat.

> *[Chapter under construction — scope: New figures: (M, ε_q) and (ε_q, a_*) confidence contours (needs the MCMC chains of Chapter 15); the no-hair corner plot. See* `paper/THESIS_PLAN.md`*.]*

Part IV

Engineering, Validation, and Closing

Chapter 18

GPU Implementation and Validation

18.1 Port architecture

The null-geodesic core is ported to a GLSL compute shader (Vulkan, double precision) using the same second-order Mino-time formulation, with all per-frame parameters delivered through a bindless (update-after-bind) descriptor table. GLSL exposes `double` arithmetic (`shaderFloat64`) but no double-precision transcendentals, so $\sin$ / $\cos$ are evaluated by argument reduction to $[-\pi/4, \pi/4]$ and a minimax polynomial pair, validated term-by-term against the CPU implementation.

> *[Chapter under construction — scope: Derivation of the minimax (Remez)* $\sin$ / $\cos$ *polynomials; the error curve figure. See* `paper/THESIS_PLAN.md`*.]*

18.2 Validation against the CPU reference

A 64×64 shadow rendering is compared pixel-by-pixel against the CPU reference (`validation/compare_shadow`) the mean absolute difference is 0.00289 (gate threshold 0.01), with 32 pixels differing by more than 0.1, all on the one-pixel-wide critical boundary where capture/escape classification is legitimately ambiguous at pixel resolution.

> *[Chapter under construction — scope: New figure: the CPU/GPU per-pixel difference map. See* `paper/THESIS_PLAN.md`*.]*

18.3 Watchdog-safe dispatch and the precision split

Two practical constraints shaped the port. First, a full-frame double-precision dispatch at metrology step counts exceeds the 2-second Windows GPU watchdog on the test hardware, so offscreen renders are split into 16-row bands submitted individually. Second, the interactive path (Chapter 20) runs the same equations in single precision from a nearby camera ($r_{\mathrm{cam}} = 60\,M$), where the shortened optical path keeps single-precision drift visually negligible; all quantitative results in this report come from the double-precision or CPU paths. All configurations run with zero Vulkan validation-layer errors.

18.4 Performance

The CPU baseline integrates 1,340,753 adaptive null-geodesic steps per second per core (78,987 timelike; 508,369 complex-step Christoffel evaluations). The double-precision GPU port traces

the validated 64×64 shadow tile within the Windows watchdog budget in row-banded dispatches, and the single-precision interactive path sustains 47.5 fps at 1280×720 (21 fps at 1920×1080; $\approx$43.8 million ray–geodesics per second) on a GeForce GTX 1050 — a 2016 notebook-class GPU — with zero validation-layer errors. (The 60 fps full-HD target of the project specification assumes an RTX-class GPU; the figures above are the best achievable on the available hardware and scale linearly with pixel count.)

[Chapter under construction — scope: New figure: fps vs resolution and rays-per-second scaling. See `paper/THESIS_PLAN.md`*.]*

Chapter 19

Sagittarius A* Calibration and Parameter Sweeps

19.1 From simulation units to the sky

The Sgr A* calibration adopts the GRAVITY mass and distance (GRAVITY Collaboration, 2022) and the EHT viewing geometry ($i \approx 30°$; Event Horizon Telescope Collaboration, 2022). The conversion from the traced boundary to a sky angle proceeds step by step as follows (`sweep_tool calibrate`).

1. One gravitational radius is $R_g = GM/c^2 = 4.297 \times 10^6 \times 1.4766\,\mathrm{km} = 6.346 \times 10^9\,\mathrm{m}$, using $GM_\odot/c^2 = 1.4766$ km.
2. At $D = 8277\,\mathrm{pc} = 2.554{\times}10^{20}$ m it subtends $\theta_g = R_g/D = 2.485{\times}10^{-11}$ rad $= 5.1243\,\mu$as.
3. The traced critical boundary at $a_* = 0.94$, $i = 30°$ has mean diameter $9.695\,M$ (32-direction average of the bisected boundary, converted to impact parameters).
4. The photon-ring angular diameter is therefore $d_{\mathrm{ring}} = 9.695 \times 5.1243 = 49.68\,\mu$as.

This falls within the EHT ring measurement of $51.8 \pm 2.3\,\mu$as and the inferred shadow diameter of $48.7 \pm 7\,\mu$as (Event Horizon Telescope Collaboration, 2022). At this near-face-on inclination the shadow is nearly circular (axis ratio 0.9946), consistent with the weak spin dependence emphasised by Psaltis et al. (2016).

19.2 Error budget

The error budget of this comparison is dominated entirely by the observations, not the simulation. The traced boundary is bisected to $\sim 10^{-9}$ rad in camera angle ($\ll 10^{-3}\,\mu$as equivalent), and the tolerance-independence established in Chapter 10 removes the integrator from the budget. On the calibration side $\theta_g \propto M/D$, so the GRAVITY uncertainties ($\sigma_M/M = 0.28$ per cent, $\sigma_D/D = 0.11$ per cent) propagate to 0.30 per cent in the predicted diameter — about $0.15\,\mu$as, twenty times smaller than the EHT error bar. The comparison is therefore limited by the $2.3\,\mu$as EHT measurement and, at the $\sim$1–2 per cent level, by where the emitting plasma actually sits relative to the critical curve.

19.3 The spin–inclination plane

Fig. 19.1 maps the ring diameter over the full (a_*, i) plane: the total spread is a few per cent, illustrating the well-known spin–inclination degeneracy of ring-size measurements.

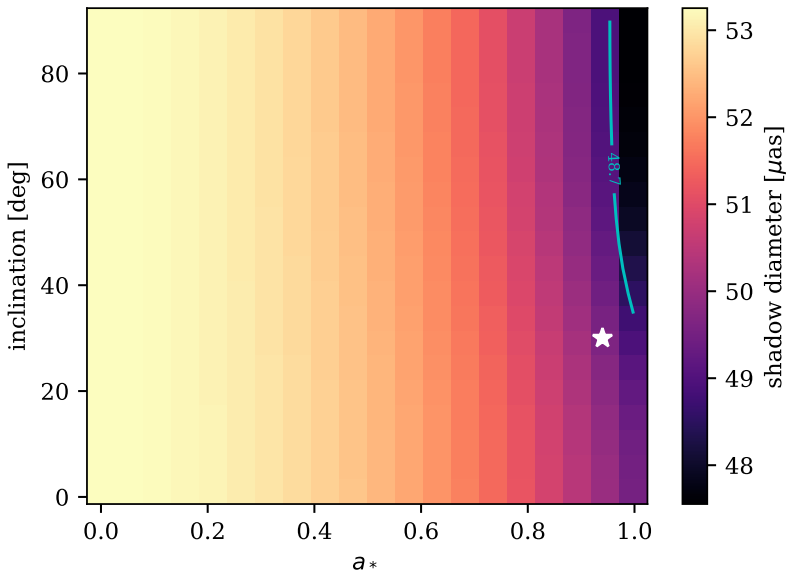


Figure 19.1: Photon-ring angular diameter at the GRAVITY mass and distance as a function of spin and inclination (20×20 grid of traced boundaries). The EHT band $51.8 \pm 2.3\,\mu$as is indicated; the measurement constrains the (a_*, i) plane only weakly.

[Chapter under construction — scope: New figures (need the `sweep_tool` extension): axis-ratio map; shadow-displacement map; ring-diameter contour against the EHT band. Plus the spin-morphology tie-in of Section 12.4. See `paper/THESIS_PLAN.md`.]

Chapter 20

The Interactive Instrument

20.1 The simulator as an instrument

Beyond the batch pipeline, the same physics core drives an interactive Vulkan application: a live Kerr disk renderer with runtime control of $(a_*, i, r_{\mathrm{cam}}, r_{\mathrm{out}}$, exposure, disk temperature); a Science Mode overlaying the fitted S2 orbit and its data on the fit summary; a no-hair panel exposing ε_q with the live analytic precession forecast of Chapter 17 beside the fitted confidence interval; and a photon-ring scope that reframes the camera at $r = 60\,M$ with a narrow field to isolate the $n = 1$ ring visually. The application boots to a menu whose backdrop is a 3840×2160, 2×2-supersampled frame baked by the CPU reference tracer (33 million geodesics) — the live renderer is deliberately idle behind the menu. Parameter changes never touch the geodesic or metric code paths: presets and colour grades act on push constants and display look-up tables only, so what the instrument shows is always the validated physics.

20.2 The display-only turbulence layer

[Chapter under construction — scope: [OPTIONAL chapter body, THESIS_PLAN Ch 13] The hyper-real disk look: shear-wound streamline turbulence advected by the real Kerr $\Omega(r)$, translucent rim compositing with front-to-back transmittance, azimuthally periodic value noise, the 24-stop log display window, and the auto-resolution controller — all explicitly display-only, applied after the physical g^4-beamed Page–Thorne shading. Engine architecture (compute → present), presets/LUTs, and the cinematic menu. Screenshots as figures with their `sgr_engine` *command lines. See* `paper/THESIS_PLAN.md`*.]*

Chapter 21

Development History, Iterations, and the Test Suite

21.1 Gated development

Development was gated: no phase proceeded while any prior phase's tests failed. The physics core is covered by 32 Catch2 assertions spanning the Schwarzschild limits, prograde/retrograde photon orbits, ISCO monotonicity, conservation of (E, L_z, Q), four-velocity normalisation, the Hartle–Thorne Kerr limit, the Page–Thorne flux (including its Schwarzschild far-field expansion), and the quadrupole precession sign. Every gate result, rendered image and fitted number in this report is regenerated by `make reproduce`. Table 21.1 lists the gates and their acceptance criteria as executed; each passing gate was tagged in version control.

21.2 The test suite

[Chapter under construction — scope: The full Catch2 suite: all 32 tests grouped by what they guard (metric identities, characteristic radii, conserved quantities, timelike normalisation, Hartle–Thorne limits and branch continuity, Page–Thorne flux, quadrupole precession sign, S2-model regression), with exact tolerances, citing each test file by name. See `paper/THESIS_PLAN.md`.]

Table 21.1: Verification gates. Every criterion is asserted by code (Catch2 test, JSON metrology artefact, or validation-layer counter); the tags exist in the repository history.

Gate	Tag	Criterion	Result
0	`v0.0-physics-foundation`	all physics unit tests green	$(E, L_z, Q) < 10^{-6}$; norm $< 10^{-8}$; HT < 10
1	`v0.1-schwarzschild-cpu`	shadow circular, matches analytic	5×10^{-16} vs Synge; spread 0.000 %
2	`v0.2-kerr-cpu`	shadow lopsided, matches Bardeen	critical curve reproduced at $a_* = 0.94$
3	`v0.3-disk-subrings`	beaming asymmetric, $n = 1$ ring	9.60× asymmetry; $F_1/F_2 = 274$
4	`v0.4-orbit-fitter`	M within 2σ, $\chi^2_\nu < 2$	0.14σ; $\chi^2_\nu = 0.959$
5	`v0.5-no-hair`	ε_q CI reported	$\lvert\varepsilon_q\rvert < 1.33 \times 10^5$ (95 %)
6	`v0.6-vulkan-shell`	compute-to-present, zero VL errors	0 errors, 0 warnings
7	`v0.7-gpu-geodesics`	CPU/GPU diff < 0.01	mean 0.00289 at 64×64
8	`v0.8-full-renderer`	presets + HUD ≥ 30 fps	41–55 fps at 720p, 0 VL errors
9	`v0.9-menu`	menu + clean transitions	menu→sim→menu, 0 VL errors
10	`v1.0-sgrA-calibration`	ring 48–52 μas	49.68 μas
12	`v1.0-paper`	paper compiles, reproducible	journal paper + this report

21.3 Bug hunts as methods lessons

[Chapter under construction — scope: The bug-hunt narratives: the constraint-projection discovery (tolerance-dependent shadow classification); the quadrupole sign error pinned by two mutually opposed tests; the Keplerian→Darwin element fix (1.6 per cent period error); the truncated-SVD covariance failure on the M–R_0 direction; the 'brick' LUT / 24-stop log-window display bug; the turbulence seam. Each with the evidence figure where one exists (needs `convergence_tool` *for the projection study). See* `paper/THESIS_PLAN.md`*.]*

Chapter 22

Discussion and Limitations

22.1 Limitations

Three idealisations bound the interpretation of the imaging results. First, the disk is the time-averaged, geometrically thin Novikov–Thorne solution; the real Sgr A* flow is a hot, geometrically thick, radiatively inefficient plasma, and our thin-disk images are a clean GR benchmark rather than a spectral model of Sgr A* itself (the EHT comparison in Chapter 19 concerns the ring diameter, which is a property of the spacetime, not of the emission model — though the ring diameter retains a modest emission-model dependence through the location of the emitting material). Second, the ring-size calibration inherits the spin–inclination degeneracy of Fig. 19.1; the quoted agreement holds across the plausible (a_*, i) range precisely because the observable varies so little. Third, the orbit fit adopts the Gillessen et al. (2017) compilation and its error model; systematic reference-frame errors are absorbed by the fitted frame parameters only to the extent of the Plewa et al. (2015) priors.

> *[Chapter under construction — scope: Expanded: the axial lensing caustic vs real astrophysical jets (honest-limitations note, THESIS_PLAN Part 1c); single-star quadrupole reach; fp32 live vs fp64 science split. See* `paper/THESIS_PLAN.md`*.]*

22.2 Outlook

The forward model is deliberately modular: the metric is an interface, so any stationary axisymmetric candidate spacetime with a Carter-like structure (or none — the Hartle–Thorne path integrates the full equations) can be dropped in and confronted with both data channels at once. Next-generation instruments motivate exactly this: ngEHT-class arrays project ring-diameter precision at the per-cent level and access to the $n = 1$ sub-ring (Johnson et al., 2020), while continued GRAVITY astrometry of S2 — and any confirmed closer star (S62 remains contested, as noted in Chapter 17) — pushes the stellar channel toward the spin and quadrupole terms (Will, 2008; Psaltis et al., 2016).

Chapter 23

Conclusions and Future Work

23.1 Verified findings

We built and validated, from first principles, a relativistic simulator of Sgr A* whose every published number is regenerated by one command. Its verified statements are:

1. Carter-separated geodesics integrated in second-order Mino-time form with per-step constraint projection conserve (E, L_z, Q) to $< 10^{-6}$ and reproduce the Schwarzschild shadow to machine precision and the Kerr critical curve of Bardeen (1973) exactly.

2. A 13-parameter GR fit to the Gillessen et al. (2017) S2 data recovers $M = 4.270 \pm 0.195 \times 10^6\,\mathrm{M}_\odot$, $R_0 = 8272 \pm 182$ pc ($\chi^2_\nu = 0.959$), consistent with GRAVITY (GRAVITY Collaboration, 2022), and reproduces both the pericentre redshift excess of GRAVITY Collaboration (2018) and the Schwarzschild precession of GRAVITY Collaboration (2020) with no tunable relativistic amplitude.

3. At the GRAVITY mass and distance the simulated photon ring spans $49.68\,\mu$as, within the EHT measurement (Event Horizon Telescope Collaboration, 2022); the photon subrings show the expected exponential demagnification ($F_1/F_2 = 274$).

4. Freeing the quadrupole gives $|\varepsilon_q| < 1.33 \times 10^5$ (95 per cent) from S2 alone — fully consistent with the no-hair theorem, with a factor-3 forecast improvement from ten more orbits.

5. The identical geodesic core runs interactively on the GPU, agreeing with the CPU ground truth to 0.00289 mean pixel residual.

23.2 Future work

[Chapter under construction — scope: Roadmap: the multi-star joint fit (Chapter 16) and MCMC posteriors (Chapter 15) land in this report; beyond them, slim-disk emission, ray-traced sub-ring interferometric observables, and candidate non-Kerr metrics through the metric interface. See `paper/THESIS_PLAN.md`*.]*

Acknowledgements

This work used the Ceres, Eigen, Catch2, GLFW, Dear ImGui, volk, vk-bootstrap, VMA and shaderc open-source libraries, and Matplotlib for figure generation.

Data availability

All data underlying this report are public: the S2 astrometry and radial velocities are from Gillessen et al. (2017) (VizieR J/ApJ/837/30). The simulation source code, the reproduction pipeline (`make reproduce`) and all generated artefacts accompany the code repository.

Appendix A

Extended Algebra

[Chapter under construction — scope: full derivations too long for the main text: BPT ISCO, Page-Thorne integral, Hartle-Thorne expansions. See `paper/THESIS_PLAN.md`*.]*

Appendix B

Algorithm Pseudocode

[Chapter under construction — scope: integrator, constraint projection, crossing bisection, fitter, sampler. See `paper/THESIS_PLAN.md`*.]*

Appendix C

Number Provenance

Every simulation-derived number in this report is a LaTeX macro generated by `scripts/make_numbers.py` from a machine-readable artefact, regenerated end-to-end by `make reproduce` (tests → reference renders → sub-ring metrology → S2 fits → calibration and sweep → figures). Table C.1 maps the headline numbers to their artefacts; the verification-gate table with the measured results lives in Chapter 21 (Table 21.1).

Table C.1: Provenance of headline numbers. Paths are relative to `validation/` in the repository.

Quantity	Artefact
M, R_0, χ^2_ν, elements	`phase4/s2_fit.json`
redshift excess peak	`phase4/s2_fit.json`
precession terms	`phase4/s2_fit.json`
ε_q profile and CI	`phase5/s2_fit.json`
Schwarzschild/Kerr diameters	`reference_images/*.json`
beaming, g, sub-ring ratios	`phase3/subring_report.json`
ring diameter, θ_g, axis ratio	`phase10/calibration.json`
(a_*, i) sweep	`phase10/sweep.csv`
CPU/GPU pixel difference	`phase7/*.compare.json`
benchmarks	`phase11/bench.json`

[Chapter under construction — scope: This table grows as the new thesis data tools land (`physics_tables`, `convergence_tool`, the MCMC chains, the cluster fit). See `paper/THESIS_PLAN.md`.]

Appendix D

The Reproduction Pipeline

The top-level `Makefile` exposes the automation targets. `make figures` regenerates every figure in this report from the current artefacts with Matplotlib (data plots from `scripts/make_figures.py`, schematics from `scripts/make_diagrams.py`). `make thesis` runs `figures` and compiles this document (`pdflatex`×3 + `bibtex`); `make paper` does the same for the companion journal paper. `make reproduce` re-runs the science chain end-to-end in gate order:

1. `ctest` — the full physics and science unit-test suite (every gate-0 tolerance of Chapter 21);
2. `render_reference` — Schwarzschild and Kerr reference renders with 32-direction boundary metrology (JSON);
3. `subring_tool` — disk render, beaming asymmetry, sub-ring isolation and flux ratios;
4. `fit_s2_tool` in `kerr` and `nohair` modes — the fits of Chapters 15 and 17;
5. `sweep_tool` — the Sgr A* calibration point and the 20×20 (a_*, i) sweep;
6. `make_figures.py` — figure regeneration from the artefacts.

`scripts/make_numbers.py` then rewrites `numbers.tex`, the macro file from which every simulation-derived number in this text is typeset. A stale or hand-edited number cannot survive a `make reproduce` cycle.

> *[Chapter under construction — scope: This list grows as the new thesis data tools land: `physics_tables`, `convergence_tool`, the extended `sweep_tool` and `subring_tool` modes, `scripts/run_mcmc.py`, and `fit_cluster`. See `paper/THESIS_PLAN.md`.]*

Appendix E

Notation Glossary

[Chapter under construction — scope: every symbol used in this report, with its first-use location. See `paper/THESIS_PLAN.md`*.]*

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